Identification and Quantification of K, Na and Mn in Textured Soy Protein Sold in the City of Rio de Janeiro Using the Argonauta Research Reactor of IEN/CNEN and the Techniques of Instrumental Neutron Activation Analysis and γ-ray Spectrometry

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1. INTRODUCTION

Soybean (Glycine max) is a legume widely cultivated in Brazil, standing out for its high nutritional value. Textured Soybean Protein, TSP, can be obtained by an industrial thermoplastic extrusion process on soybean flour.

According to the available literature, it is known that each cultivar of soybean absorbs properties from the soil, as well as from fertilizers and added manures, and in this way, they become foods rich in essential minerals with nutritional relevance such as potassium (K), sodium (Na), and manganese (Mn).

A diet rich in potassium contributes to the control of blood pressure, the prevention of cardiovascular diseases and kidney disorders, and the improvement of overall health. The Institute of Medicine (IOM) suggests an adequate intake of 4700 mg/ day for adults. Sodium is important for maintaining acid-base balance, ensuring the body's pH remains within a healthy range. The World Health Organization (WHO) recommends a daily intake of 2 g. Manganese, when consumed in controlled doses, supports various physiological functions; However, excessive exposure can lead to neurological disorders. The adequate intake for adults is estimated to range from 1.8 mg/ day for women to 2.3 mg/ day for men.

In this context, the present study aimed to identify and quantify the concentrations of these elements in a commercial sample of TSP. For this purpose, the techniques of Instrumental Neutron Activation Analysis (INAA), and Gamma-Ray Spectrometry (GS) were employed, using a high-purity germanium detector, HPGe. The neutron source used was the Argonauta research reactor, located at the Instituto de Engenharia Nuclear (IEN/CNEN), operating under conditions of 34 W power and a thermal neutron flux of 10^8 n/cm²·s for 30 minutes.

2. MATERIALS AND METHODS

The TSP samples were converted into a fine powder and placed into polyethylene vials. Their masses were determined in a four-decimal precision balance. The reference reagent samples (KCl, NaCl and MnO_2) were also placed on individual vials and weighed.

Among neutron sources, nuclear reactors provide the highest analytical sensitivity for most elements due to intense neutron fluxes. 95% of the total incident neutron flux is composed of thermal neutrons (E<0,5eV), which are primarily responsible for inducing radiative capture reactions (n, γ) that form the basis of the activation techniques used in the study.

INAA is non-destructive technique for elemental quantification based on radiative neutron capture (n,γ), using high-resolution detectors such as High-Purity Germanium (HPGe). Quantitative analysis is performed by the comparative method, in which target and reference samples are irradiated simultaneously under identical conditions. The concentrations in the target sample are then determined by directly comparing their gamma spectrometric results with those of the reference standards.

GS is a technique used for the identification and quantification of radionuclides through the detection of emitted gamma radiation. It produces energy spectra, which are graphs relating the number of events (counts) and the energy of the detected photons (keV). By analyzing the graphs, is possible to identify the elements present in the target sample, as the radioisotopes are characterized by their half-lives and by the energy of the radiation they emit. To avoid false diagnoses that may lead to incorrect identifications of the present elements, the half-life test is used. It consists in the analysis of successive spectra acquired from the same source and observing the decrease of the areas in the graph.

3. RESULTS

Table 1: Half-Life Test

TSP	γ Energy (keV)	$T_{1/2}$ (IAEA) (s)	$T_{1/2}$ (this work) (s)
^{42}K	1,524.86	44,478.0	45,123.2±720.5
^{24}Na	1,368.62	53,841.6	51,233.4±3196.8
^{56}Mn	846.76	9,284.0	9,430.6±187.9

Table 2: INAA Comparative Method

TSP	C_{PTS} (Hamdy) (mg/100 g)	$\overline{C_{PTS}}$ (this work) (mg/100 g)
K	2,090.0	(1,520.7±93.3)
Na	0.9	(4.3±1.4)
Mn	3.8	(2.3±0.1)

4. CONCLUSIONS

The results demonstrate the presence of the target elements, as well as their concentration values. Regarding the comparison of the studies, several factors should be taken in consideration: The geographic origin of the sample, soil composition, agricultural practices and the extrusion process can influence the elemental composition of soy-based products.

It is concluded that the INAA and GS techniques proved to be effective both in the identification and quantification of the elements present in the analyzed sample. The combined use of these techniques offers significant advantages, including non-destructive analysis, high sensitivity, multielement capability, and minimal sample preparation, enabling efficient utilization of the infrastructure available at the Argonauta research reactor.

The results indicated the analyzed TSP is an excellent source of potassium, have a low concentration of sodium and an adequate concentration of manganese. Thus, the data obtained reinforce the nutritional value of TSP, not only as a plant-based protein source but also as a functional food that can significantly contribute to a balanced diet.

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